Andrew Dai

AI + Full-Stack Developer | 10 Consecutive Hackathon wins

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Education

University of Waterloo — Expected May 2031

Bachelor of Computer Science with Minor in Physics — *Waterloo, ON*

Experience

JAMHacks — Aug. 2025 – Sep. 2026

Head Organizer, Lead Software Engineer, Judge Panelist — *Waterloo, ON*

- Built jamhacks.ca using **React + Node.js + MongoDB**, serving **12,000+ monthly visits** and **1,000+ registered users**
- Scaled Canada's largest high school hackathon, increasing **attendees by 41.5%** in under a year on a **$15,000 budget**
- Coordinated and worked with engineers from Google, Meta, Apple, & Microsoft to judge **100+ student-built projects**
- Reviewed **576 commits** for clarity and maintainability, providing code review feedback to a **team of 5 engineers**

Cullergrader (Open Source) — May. 2024 – Present

Developer & Maintainer — *Toronto, ON*

- Built an open source tool to bulk sort similar photos by **perceptual hash**, designed to handle large folders of **100GB+**
- Generates **24-byte hashes** in LAB colour space per photo, using hash hamming distance to compute visual similarity
- Implemented configurable multi-threading to process large batches **up to 64x faster** than single-threaded alternatives

Hackathons — Dec. 2022 – Present

Competitor, Judge Panelist, Mentor — *Canada*

- Won **10 consecutive hackathons** out of 18 attended, shipping and pitching full-stack products in **under 24 hours**
- Awarded **$4,000+ in prizes** for technical solutions built on **Databricks, MongoDB, Cohere, & Roboflow** APIs
- Finalist for **Major League Hacking's** annual **Top 50 award**, an international organization of **150,000+ members**

Projects

Semantic Species Retrieval – *OpenAI, Google AI Platform, Flask, Next.js, SQLite*

- A scalable animal identification pipeline that supports 9,641 species, **366% more than Google SpeciesNet's 2,066**
- **Eliminated per-species costs** by mapping visual traits to embeddings, new species can be added without extra training
- Published an animal-information dataset covering 9,641 species on **HuggingFace**, earning over **100 downloads**

Agentic Workflow Prototyper – *Google AI Platform, Moorcheh.ai RAG, Flask, MongoDB, Next.js, Supabase*

- Built a **tool-based agentic system** to build executable API workflows using a custom drag-and-drop node system
- Integrated Moorcheh.ai **high-speed RAG** for **vector retrieval 6.6x faster** than typical Pinecone and Cohere pipelines

Physics-Informed Shuttlecock Tracking – *YOLOv8, LiDar, Roboflow, Next.js, Flask*

- Tracks fast-moving small objects (**0.024% of frame area**) in 3D space using **motion isolation** and **trajectory modeling**
- Designed a rudimentary **real-time world model** with gyroscope-calculated gravity data and **live court segmentation**
- Combined LiDar sensors with **depth estimation models** for reliable spatial positioning with respect to camera extrinsics

Skills

- **Languages:** Python, JavaScript, TypeScript, HTML, CSS, C++, C#, Java, Racket
- **Frameworks & Libraries:** Flask, React, Next.js, Tailwind, PyTorch, NumPy, Pandas, OpenCV, LangChain, MongoDB
- **Development Tools:** Git, Claude Code, Codex, GCP, AWS, Docker, Terraform, Postman